

# ESP32-S3-Touch-LCD-1.47

From Waveshare Wiki

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## Overview

[[Expand](#)]

## Usage Instructions

ESP32-S3-Touch-LCD-1.47 currently provides two development tools and frameworks, **Arduino IDE** and **ESP-IDF**, providing flexible development options, you can choose the right development tool according to your

### ESP32-S3-Touch-LCD-1.47



(<https://www.waveshare.com/esp32-s3-touch-lcd-1.47.htm>)

172×320, I2C / SPI

project needs and personal habits.

## Development Tools

### Arduino IDE

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(/wiki/File:270px-Arduino-IDE-logo.jpg)

Arduino IDE is an open source electronic prototyping platform, convenient and flexible, easy to get started. After a simple learning, you can start to develop quickly. At the same time, Arduino has a large global user community, providing an abundance of open source code, project examples and tutorials, as well as rich library resources, encapsulating complex functions, allowing developers to quickly implement various functions.

### ESP-IDF

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(/wiki/File:270px-ESP-IDF-logo.jpg)

ESP-IDF, or full name Espressif IDE, is a professional development framework introduced by Espressif Technology for the ESP series chips. It is developed using the C language, including a compiler, debugger, and flashing tool, etc., and can be developed via the command lines or through an integrated development environment (such as Visual Studio Code with the Espressif IDF plugin). The plugin offers features such as code navigation, project management, and debugging, etc.

Each of these two development approaches has its own advantages, and developers can choose according to their needs and skill levels. Arduino are suitable for beginners and non-professionals because they are easy to learn and quick to get started. ESP-IDF is a better choice for developers with a professional background or high performance requirements, as it provides more advanced development tools and greater control capabilities for the development of complex projects.

## Components Preparation

- ESP32-S3-Touch-LCD-1.47 x1
- TF card x 1 (32GB and below)

- Card reader x1

Before operating, it is recommended to browse the table of contents to quickly understand the document structure. For smooth operation, please read the FAQ carefully to understand possible problems in advance. All resources in the document are provided with hyperlinks for easy download.

## Working with Arduino

This chapter introduces setting up the Arduino environment, including the Arduino IDE, management of ESP32 boards, installation of related libraries, program compilation and downloading, as well as testing demos. It aims to help users master the development board and facilitate secondary development.

### Environment Setup

#### Download and Install Arduino IDE

- Click to visit the Arduino official website (<https://www.arduino.cc/en/software>), select the corresponding system and system bit to download



 **Arduino IDE 2.3.3**

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger.

For more details, please refer to the [Arduino IDE 2.0 documentation](#).

Nightly builds with the latest bugfixes are available through the section below.

**SOURCE CODE**

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

**DOWNLOAD OPTIONS**

- Windows** Win 10 and newer, 64 bits
- Windows** MSI installer
- Windows** ZIP file
- Linux** AppImage 64 bits (X86-64)
- Linux** ZIP file 64 bits (X86-64)
- macOS** Intel, 10.15: "Catalina" or newer, 64 bits
- macOS** Apple Silicon, 11: "Big Sur" or newer, 64 bits

[Release Notes](#)

(/wiki/File:ESP32-S3-AMOLED-1.91-Ar-software-01.png)

- Run the installer and install all by default

## Install ESP32 Development Board

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- Before using ESP32-related motherboards with the Arduino IDE, you must first install the software package for the **esp32 by Espressif Systems** development board
- According to **board installation requirement**, it is generally recommended to use **Install Online**. If online installation fails, use **Install Offline**.
- For the installation tutorial, please refer to Arduino board manager tutorial ([https://www.waveshare.com/wiki/Arduino\\_Board\\_Managers\\_Tutorial](https://www.waveshare.com/wiki/Arduino_Board_Managers_Tutorial))
- ESP32-S3-Touch-LCD-1.47 required development board installation description

| Board name                 | Board installation requirement       | Version number requirement |
|----------------------------|--------------------------------------|----------------------------|
| esp32 by Espressif Systems | "Install Offline" / "Install Online" | ≥3.0.0                     |

## Install Library

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- When installing Arduino libraries, there are usually two ways to choose from: **Install online** and **Install offline**. **If the library installation requires offline installation, you must use the provided library file**  
For most libraries, users can easily search and install them through the online library manager of the Arduino software. However, some open-source libraries or custom libraries are not synchronized to the Arduino Library Manager, so they cannot be acquired through online searches. In this case, users can only manually install these libraries offline.
- For library installation tutorial, please refer to Arduino library manager tutorial ([https://www.waveshare.com/wiki/Arduino\\_Library\\_Manager\\_Tutorial](https://www.waveshare.com/wiki/Arduino_Library_Manager_Tutorial))
- **ESP32-S3-Touch-LCD-1.47 library file path:**

```
..\ESP32-S3-Touch-LCD-1.47-Demo\Arduino\libraries
```

■ ESP32-S3-Touch-LCD-1.47 Library file installation instructions

| Library Name            | Description          | Version | Library Installation Requirement                            |
|-------------------------|----------------------|---------|-------------------------------------------------------------|
| lvgl                    | Graphical library    | v8.4.0  | "Install Online" (requires copying the demos folder to src) |
| GFX_Library_for_Arduino | LCD driver library   | v1.5.9  | "Install Online"                                            |
| esp_lcd_touch_axs5106l  | Touch driver library | --      | "Install Offline"                                           |

## Run the First Arduino Demo

If you are just getting started with ESP32 and Arduino, and you don't know how to create, compile, flash, and run Arduino ESP32 programs, then please expand and take a look. Hope it can help you!

[Expand]

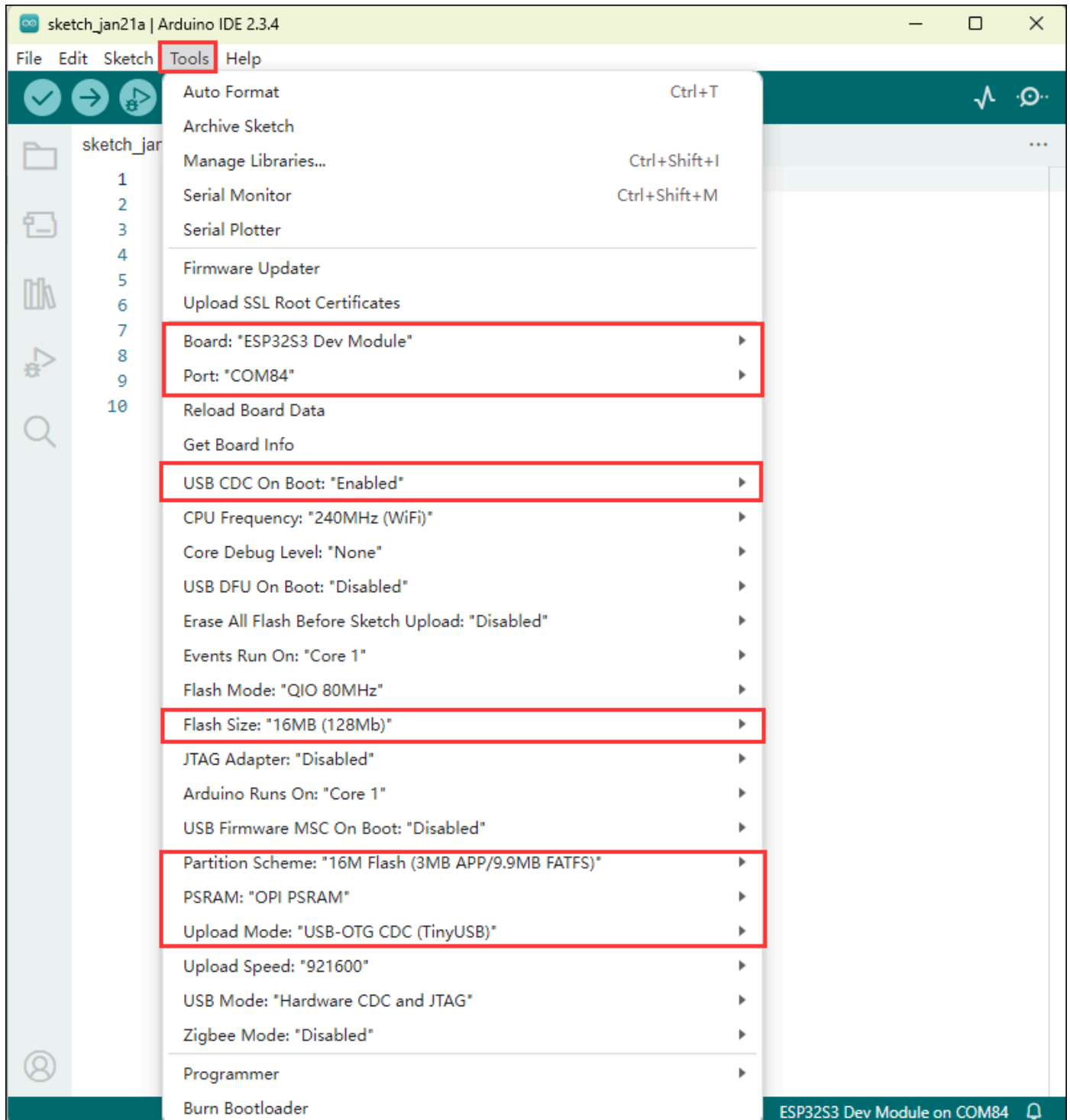
## Demo

■ ESP32-S3-Touch-LCD-1.47 Demo

| Demo               | Basic Description                                             | Dependency Library                                    |
|--------------------|---------------------------------------------------------------|-------------------------------------------------------|
| 01_gfx_helloworld  | HelloWorld is displayed on the screen                         | GFX_Library_for_Arduino                               |
| 02_sd_card_test    | Test TF card                                                  | --                                                    |
| 03_lvgl_arduino_v8 | lvgl demos                                                    | lvgl, GFX_Library_for_Arduino, esp_lcd_touch_axs5106l |
| 04_lvgl_battery    | Use the LVGL library to display the battery voltage           | lvgl, GFX_Library_for_Arduino, esp_lcd_touch_axs5106l |
| 05_lvgl_brightness | Use the LVGL library to control and display screen brightness | lvgl, GFX_Library_for_Arduino, esp_lcd_touch_axs5106l |
| 06_lvgl_image      | Use the LVGL library to display images                        | lvgl, GFX_Library_for_Arduino, esp_lcd_touch_axs5106l |

## Arduino Project Parameter Setting

- Each demo must be configured with parameters before flashing



(/wiki/File:ESP32-S3-Touch-LCD-1.47-Demo-02.png)

## 01\_gfx\_helloworld

**【Demo description】**

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 uses the `GFX_Library_for_Arduino` library to drive the screen and display HelloWorld on the screen

### 【Hardware connection】

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- Connect the board to the computer

### 【Code analysis】

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- Create an object bus of the `Arduino_ESP32SPI` class to configure the SPI bus GPIO, and create an object gfx of the `Arduino_ST7789` class to drive the ST7789 display

```
Arduino_DataBus *bus = new Arduino_ESP32SPI(45 /* DC */, 21 /* CS */, 38 /* SCK */,
39 /* MOSI */);

Arduino_GFX *gfx = new Arduino_ST7789(
    bus, 47 /* RST */, 0 /* rotation */, false /* IPS */,
    172 /* width */, 320 /* height */,
    34 /*col_offset1*/, 0 /*uint8_t row_offset1*/,
    34 /*col_offset2*/, 0 /*row_offset2*/);
```

### 【Result demonstration】

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- HelloWorld is displayed on the screen.



LCD-1.47-Demo-03.jpg)

(/wiki/File:600px-ESP32-S3-Touch-

## 02\_sd\_card\_test

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### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 tests the read and write functions of the TF card

### 【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot (**TF card needs to be formatted as FAT32**)



## 【Code analysis】

### ■ TF card initialization

```
if(! SD_MMC.setPins(clk, cmd, d0, d1, d2, d3)){
    Serial.println("Pin change failed!");
    return;
}
if (!SD_MMC.begin()) {
    Serial.println("Card Mount Failed");
    return;
}
```

## 【Result demonstration】

- Insert the TF card into the computer, and you can find two more files: test.txt and foo.test. The content of the foo.txt is Hello World!, and the content of the test.txt is empty

 foo.txt test.txt

(/wiki/File:ESP32-S3-Touch-LCD-1.47-Demo-

04.png)

## 03\_lvgl\_arduino\_v8

### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 runs the lvgl demos

### 【Hardware connection】

- Connect the board to the computer

## 【Precautions】

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- If the lvgl library is installed online, you need to copy the demos folder to src

## 【Code analysis】

---

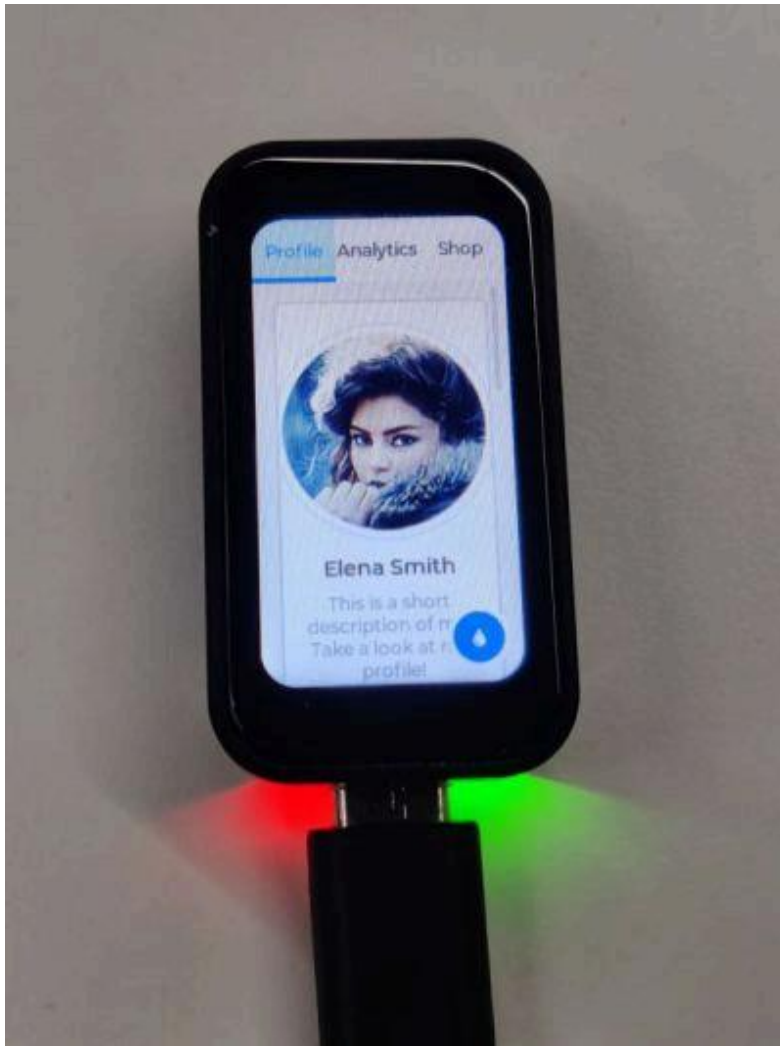
- Select the lvgl demos to run

```
lv_demo_widgets();  
// lv_demo_benchmark();  
// lv_demo_keypad_encoder();  
// lv_demo_music();  
// lv_demo_stress();
```

## 【Result demonstration】

---

- It can be operated by touch



LCD-1.47-Demo-05.jpg)

(/wiki/File:600px-ESP32-S3-Touch-

## 04\_lvgl\_battery

---

### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 displays battery voltage and ADC values on the screen using the lvgl library

### 【Hardware connection】

---

- Connect the board to the computer

### 【Code analysis】

- Initialize the UI and create a 1000ms timer to obtain the data of the adc and convert the data into a voltage

```
lvgl_battery_ui_init(lv_scr_act());
```

### 【Result demonstration】



LCD-1.47-Demo-06.jpg)

(/wiki/File:600px-ESP32-S3-Touch-

## 05\_lvgl\_brightness

### 【Demo description】

[[Collapse](#)]

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 displays the screen brightness on the screen using the lvgl library, and controls the screen brightness through the slider

### **【Hardware connection】**

---

- Connect the board to the computer

### **【Code analysis】**

---

- Initialize the UI, and create a slider value change callback, when the value of the slider changes, modify the screen brightness

```
lvgl_brightness_ui_init(lv_scr_act());
```

### **【Result demonstration】**

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(/wiki/File:600px-ESP32-S3-Touch-

LCD-1.47-Demo-07.jpg)

## 06\_lvgl\_image

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### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 runs the lvgl to display images.

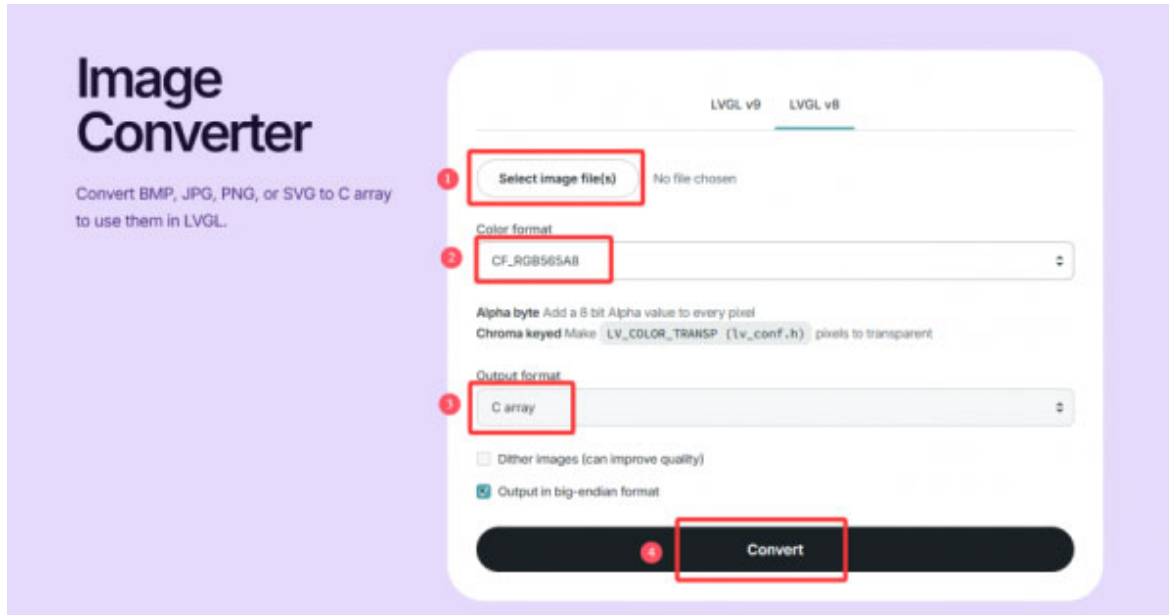
### 【Hardware connection】

---

- Connect the board to the computer

## 【Preparation】

- Open lvgl Image Converter website (<https://lvgl.io/tools/imageconverter>)
- Import the image images/image\_1.jpg and convert it, the image\_1.c file will be generated



(/wiki/File:900px-ESP32-S3-Touch-LCD-1.47-Demo-01.png)

- Copy image\_1.c to the project folder
- Add the following code to 06\_lvgl\_image.ino to declare the image

```
LV_IMG_DECLARE(image_1);
```

- Set the image to be displayed

```
lv_img_set_src(img, &image_1);
```

## 【Result demonstration】



LCD-1.47-Demo-08.jpg)

(/wiki/File:600px-ESP32-S3-Touch-

## Working with ESP-IDF

This chapter introduces setting up the ESP-IDF environment setup, including the installation of Visual Studio and the Espressif IDF plugin, program compilation, downloading, and testing of demos, to assist users in mastering the development board and facilitating secondary development.

### Environment Setup

#### Download and Install Visual Studio

- Open the download page of VScode official website (<https://code.visualstudio.com/download>), choose the corresponding system and system bit to download



# Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



↓ **Windows**

Windows 10, 11

User Installer x64 Arm64

System x64 Arm64

Installer x64 Arm64

.zip x64 Arm64

CLI x64 Arm64



↓ **.deb**

Debian, Ubuntu

↓ **.rpm**

Red Hat, Fedora, SUSE

.deb x64 Arm32 Arm64

.rpm x64 Arm32 Arm64

.tar.gz x64 Arm32 Arm64

Snap Snap Store

CLI x64 Arm32 Arm64



↓ **Mac**

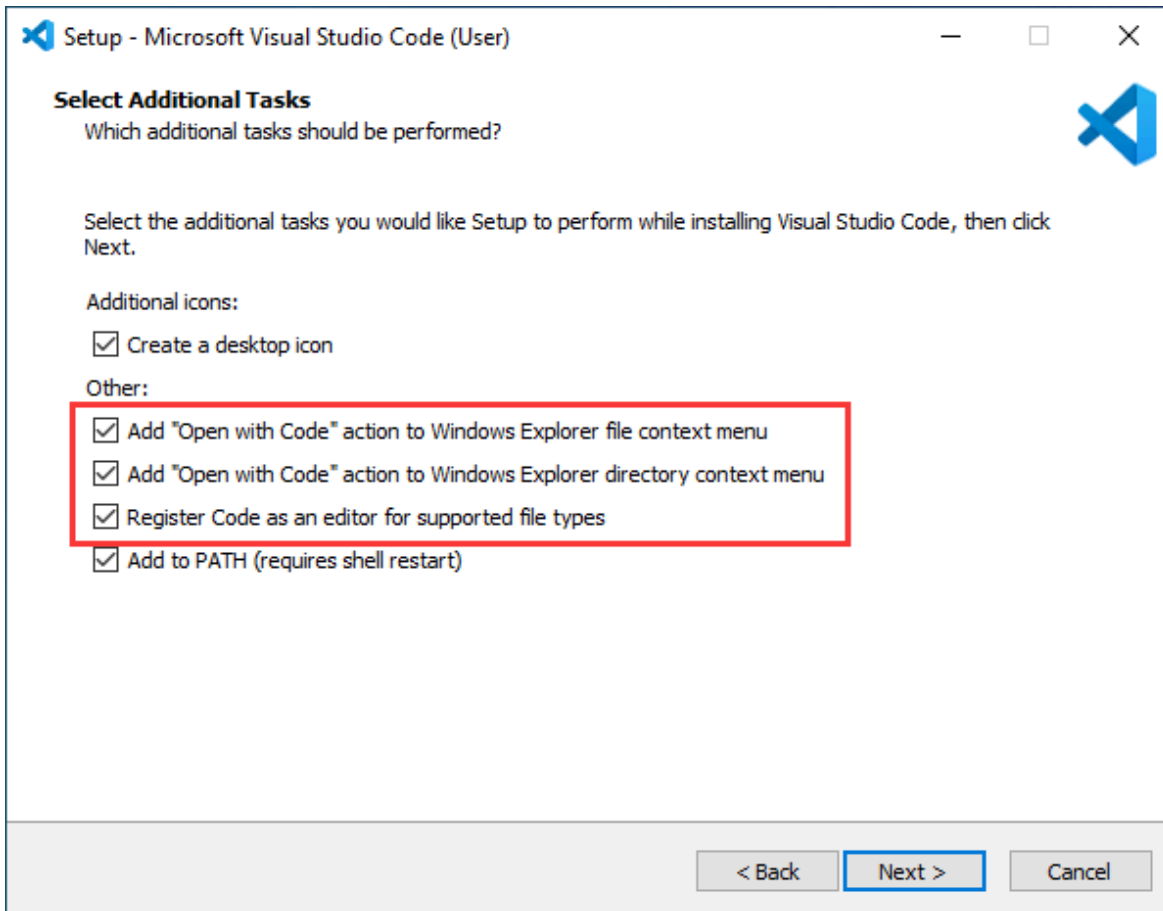
macOS 10.15+

.zip Intel chip Apple silicon Universal

CLI Intel chip Apple silicon

(/wiki/File:ESP32-S3-AMOLED-1.91-VScode-01.png)

- After running the installation package, the rest can be installed by default, but here for the subsequent experience, it is recommended to check boxes 1, 2, and 3



(/wiki/File:ESP32-S3-AMOLED-1.91-VScode-02.png)

- After the first two items are enabled, you can open VSCode directly by right-clicking files or directories, which can improve the subsequent user experience
- After the third item is enabled, you can select VSCode directly when you choose how to open it

The environment setup is carried out on the Windows 10 system, Linux and Mac users can access ESP-IDF environment setup (<https://docs.espressif.com/projects/esp-idf/en/v5.1.4/esp32s3/get-started/windows-setup.html>) for reference

## Install Espressif IDF Plugin

- It is generally recommended to use **Install Online**. If online installation fails due to network factor, use **Install Offline**
- For more information about how to install the Espressif IDF plugin, see Install Espressif IDF Plugin ([https://www.waveshare.com/wiki/Install\\_Espressif\\_IDF\\_Plugin\\_Tutorial](https://www.waveshare.com/wiki/Install_Espressif_IDF_Plugin_Tutorial))

## Run the First ESP-IDF Demo

If you are just getting started with ESP32 and ESP-IDF, and you don't know how to create, compile, flash, and run ESP-IDF ESP32 programs, then please expand and take a look. Hope it can help you!

[Expand]

Demo

- ESP32-S3-Touch-LCD-1.47 Demo

| Demo             | Basic Description                          |
|------------------|--------------------------------------------|
| 01_factory       | Comprehensive test demo                    |
| 02_sd_card_test  | Test the read and write of the TF card     |
| 03_lvgl_example  | lvgl demos                                 |
| 04_lvgl_image    | Use lvgl library to display images         |
| 05_lvgl_sd_image | Use lvgl library to display TF card images |

01\_factory

【Demo description】

[Collapse]

This demo tests the functionality of the ESP32-S3-Touch-LCD-1.47 onboard modules, the information of each module will be displayed on the screen, and the user can swipe left and right through the touch screen to switch between pages

【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot

【Code analysis】

- Peripheral initialization

```
i2c_bus_handle = bsp_i2c_init();
bsp_battery_init();
bsp_wifi_init("WSTEST", "waveshare0755");
bsp_display_init(&io_handle, &panel_handle, EXAMPLE_LCD_H_RES * EXAMPLE_LCD_DRAW
_BUFF_HEIGHT);
bsp_touch_init(&touch_handle, i2c_bus_handle, EXAMPLE_LCD_H_RES, EXAMPLE_LCD_V_R
ES, EXAMPLE_DISPLAY_ROTATION);
bsp_sdcard_init();
ESP_ERROR_CHECK(app_lvgl_init());
lv_fs_fatfs_init();

bsp_display_brightness_init();
bsp_display_set_brightness(100);
```

#### ■ LVGL graphics library configuration

```
if (lvgl_port_lock(0))
{
    lvgl_ui_init();
    lvgl_port_unlock();
}
```

### 【Result demonstration】



(/wiki/File:300px-ESP32-S3-Touch-LCD-1.47-Demo-11.jpg)



(/wiki/File:300px-ESP32-S3-Touch-LCD-1.47-Demo-12.jpg)



(/wiki/File:300px-ESP32-S3-Touch-LCD-1.47-Demo-13.jpg)



(/wiki/File:300px-ESP32-S3-Touch-LCD-1.47-Demo-14.jpg)

## 02\_sd\_card\_test

---

### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 tests the read and write functions of the TF card

### 【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot **(TF card needs to be formatted as FAT32)**

### 【Code analysis】

- TF card initialization and mounting

```

    sdmmc_slot_config_t slot_config = SDMMC_SLOT_CONFIG_DEFAULT();
#if EXAMPLE_IS_UHS1
    slot_config.flags |= SDMMC_SLOT_FLAG_UHS1;
#endif

    // Set bus width to use:
#ifdef CONFIG_EXAMPLE_SDMMC_BUS_WIDTH_4
    slot_config.width = 4;
#else
    slot_config.width = 1;
#endif

    // On chips where the GPIOs used for SD card can be configured, set them in
    // the slot_config structure:
#ifdef CONFIG_SOC_SDMMC_USE_GPIO_MATRIX
    slot_config.clk = CONFIG_EXAMPLE_PIN_CLK;
    slot_config.cmd = CONFIG_EXAMPLE_PIN_CMD;
    slot_config.d0 = CONFIG_EXAMPLE_PIN_D0;
#endif
#ifdef CONFIG_EXAMPLE_SDMMC_BUS_WIDTH_4
    slot_config.d1 = CONFIG_EXAMPLE_PIN_D1;
    slot_config.d2 = CONFIG_EXAMPLE_PIN_D2;
    slot_config.d3 = CONFIG_EXAMPLE_PIN_D3;
#endif // CONFIG_EXAMPLE_SDMMC_BUS_WIDTH_4
#endif // CONFIG_SOC_SDMMC_USE_GPIO_MATRIX

    // Enable internal pullups on enabled pins. The internal pullups
    // are insufficient however, please make sure 10k external pullups are
    // connected on the bus. This is for debug / example purpose only.
    slot_config.flags |= SDMMC_SLOT_FLAG_INTERNAL_PULLUP;

    ESP_LOGI(TAG, "Mounting filesystem");
    ret = esp_vfs_fat_sdmmc_mount(mount_point, &host, &slot_config, &mount_config, &
card);

```

## 【Result demonstration】

- Insert the TF card into the computer, and you can find three more files: test.txt, FOO.TXT and NIHAO.TXT. Among them, the content of the FOO.TXT is Hello SD!, the content of the NIHAO.TXT is Hello SD!, and the content of the test.txt is empty

 FOO.TXT NIHAO.TXT

(/wiki/File:ESP32-S3-Touch-LCD-1.47-Demo-

 test.txt

09.png)

## 03\_lvgl\_example

---

### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 runs the lvgl demos

### 【Hardware connection】

- Connect the board to the computer

### 【Code analysis】

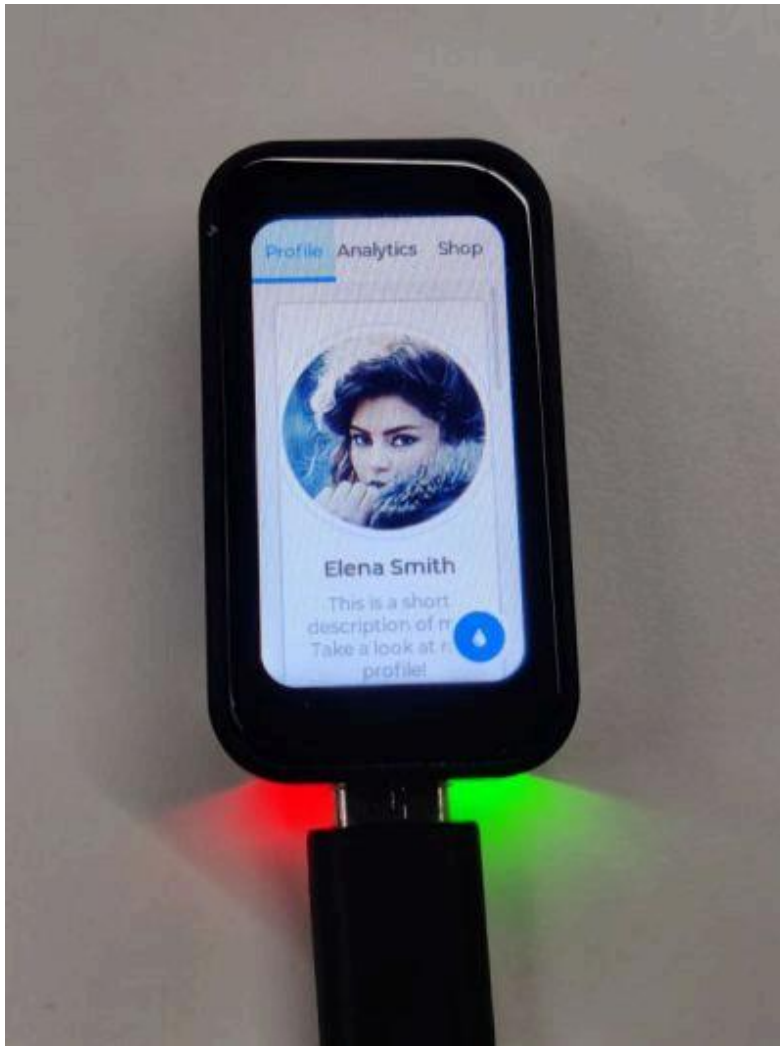
- Select the lvgl demos to run

```
// lv_demo_benchmark();  
// lv_demo_music();  
lv_demo_widgets();  
lvgl_port_unlock();
```

### 【Result demonstration】

- It can be operated by touch





LCD-1.47-Demo-05.jpg)

(/wiki/File:600px-ESP32-S3-Touch-

## 04\_lvgl\_image

---

### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates that ESP32-S3-Touch-LCD-1.47 runs the lvgl to display images.

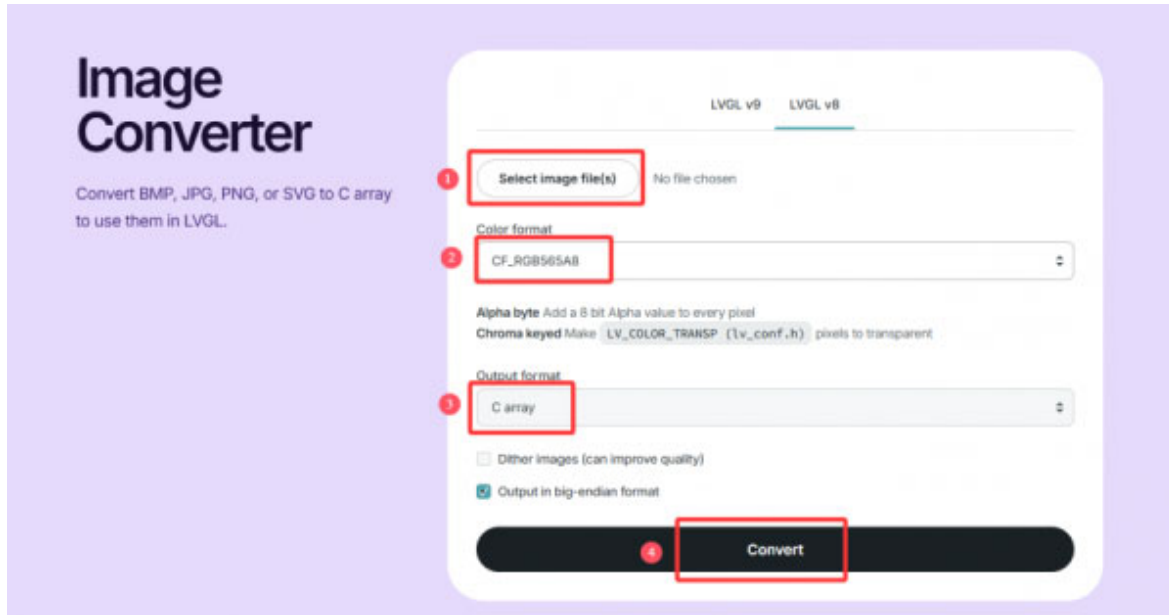
### 【Hardware connection】

---

- Connect the board to the computer

## 【Preparation】

- Open lvgl Image Converter website (<https://lvgl.io/tools/imageconverter>)
- Import the image images/image\_1.jpg and convert it, the image\_1.c file will be generated



(/wiki/File:900px-ESP32-S3-Touch-LCD-1.47-Demo-01.png)

- Copy image\_1.c to the main folder
- Add "image\_1.c" to the CMakeLists.txt in the main folder

```
idf_component_register(SRCS "main.c" "image_1.c"
                      INCLUDE_DIRS ".")
```

- Add the following code to main.c to declare the image

```
LV_IMG_DECLARE(image_1);
```

- Set the image to be displayed

```
lv_img_set_src(img, &image_1);
```

## 【Result demonstration】



LCD-1.47-Demo-08.jpg)

(/wiki/File:600px-ESP32-S3-Touch-

## 05\_lvgl\_sd\_image

---

### 【Demo description】

[\[Collapse\]](#)

This demo demonstrates how ESP32-S3-Touch-LCD-1.47 uses lvgl to display images from a TF card, supporting formats such as jpg, bmp, png, bin, etc.

### 【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot

## 【Preparation】

---

- Insert the TF card into the computer
- Copy the images folder of this project to the root directory of the TF card
- Insert the TF card into ESP32-S3-Touch-LCD-1.47

## 【Result demonstration】

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(/wiki/File:600px-ESP32-S3-Touch-

LCD-1.47-Demo-10.jpg)

# Flash Firmware Flashing and Erasing

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- **The current demo provides test firmware, which can be used to test whether the onboard device functions properly by directly flashing the test firmware**
- **bin file path:**

```
..\ESP32-S3-Touch-LCD-1.47-Demo\Firmware
```

Flash firmware flashing and erasing ([https://www.waveshare.com/wiki/Flash\\_Firmware\\_Flashing\\_and\\_Erasing](https://www.waveshare.com/wiki/Flash_Firmware_Flashing_and_Erasing)) for reference

## Resources

### Schematic Diagram

- ESP32-S3-Touch-LCD-1.47 Schematic diagram (<https://files.waveshare.com/wiki/ESP32-S3-Touch-LCD-1.47/ESP32-S3-Touch-LCD-1.47-Schematic.pdf>)

### Demo

- ESP32-S3-Touch-LCD-1.47 Demo (<https://files.waveshare.com/wiki/ESP32-S3-Touch-LCD-1.47/ESP32-S3-Touch-LCD-1.47-Demo.zip>)

### Project Document

- ESP32-S3-Touch-LCD-1.47 2D/3D file (<https://files.waveshare.com/wiki/ESP32-S3-Touch-LCD-1.47/ESP32-S3-Touch-LCD-1.47-2D3D.zip>)

### Datasheets

## ESP32-S3

- ESP32-S3 Technical Reference Manual ([https://files.waveshare.com/wiki/common/Esp32-s3\\_technical\\_reference\\_manual\\_en.pdf](https://files.waveshare.com/wiki/common/Esp32-s3_technical_reference_manual_en.pdf))

- ESP32-S3 Series Datasheet ([https://files.waveshare.com/wiki/common/Esp32-s3\\_datasheet\\_en.pdf](https://files.waveshare.com/wiki/common/Esp32-s3_datasheet_en.pdf))

## Software Tools

### Arduino

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- Arduino IDE Official download link (<https://www.arduino.cc/en/software/>)
- ESP32-Arduino official documentation (<https://docs.espressif.com/projects/arduino-esp32/en/latest/index.html>)
- Arduino-ESP32 offline component package ([https://drive.google.com/drive/folders/1Pcs\\_A4FKWvdSHnz9IEBYqOpr-noTMblv?usp=sharing](https://drive.google.com/drive/folders/1Pcs_A4FKWvdSHnz9IEBYqOpr-noTMblv?usp=sharing))

### VScode

---

- VScode official website (<https://code.visualstudio.com/download>)

### Firmware Flashing Tool

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- Flash\_download\_tool ([https://dl.espressif.com/public/flash\\_download\\_tool.zip](https://dl.espressif.com/public/flash_download_tool.zip))

## Other Resource Links

- ESP32-Arduino official documentation (<https://docs.espressif.com/projects/arduino-esp32/en/latest/index.html>)
- LVGL official documentation (<https://docs.lvgl.io/master/intro/introduction/index.html>)

## Project Resources

This section features third - party project resources. We merely provide links and bear no responsibility for content updates or maintenance. Thank you for your understanding.

### Volos Projects-Is This the Cheapest ESP32 Touchscreen Board?... Here's What I Built!



Is This the Cheapest ESP32 Touchscre  
Volos Projects



Watch on

- Youtube : <https://www.youtube.com/watch?v=fVWWXJ9sgbM> (<https://www.youtube.com/watch?v=fVWWXJ9sgbM>)

- Github : <https://github.com/VolosR/Waveshare147Touch> (<https://github.com/VolosR/Waveshare147Touch>)

## Volos Projects-Blink Sketch on Steroids 2.0 — Now With Touch Control!



### Blink Sketch on Steroids 2.0 — Now Wi

Volos Projects



Watch on

- Youtube : <https://www.youtube.com/watch?v=GevrPdXXQZY> (<https://www.youtube.com/watch?v=GevrPdXXQZY>)
- Github : <https://github.com/VolosR/Wave147moreExamples> (<https://github.com/VolosR/Wave147moreExamples>)

## Sayed Ahmadreza Razian, PhD-TinyCryptoFlow

- Github: <https://github.com/ahmadrezarazian/Waveshare-ESP32-S3-LCD1.47-TinyCryptoFlow> (<https://github.com/ahmadrezarazian/Waveshare-ESP32-S3-LCD1.47-TinyCryptoFlow>)

## FAQ

**Question: What should I do if the Arduino demo does not show up after downloading?**

**Answer:**

- Set Arduino Project Parameters ([https://www.waveshare.com/wiki/ESP32-S3-Touch-LCD-1.47#Arduino\\_Project\\_Parameter\\_Setting](https://www.waveshare.com/wiki/ESP32-S3-Touch-LCD-1.47#Arduino_Project_Parameter_Setting)) .

**Question: After the module downloads the demo and re-downloads it, why sometimes it can't connect to the serial port or the flashing fails?**

**Answer:**

- Long press the BOOT button, plug in the USB at the same time, then release the BOOT button, at this time the module can enter the download mode, which can solve most of

the problems that can not be downloaded.

**Question: Why does the module keep resetting and flicker when viewed the recognition status from the device manager?**

**Answer:**

- It may be due to Flash blank and the USB port is not stable, you can long-press the BOOT button, press RESET at the same time, and then release RESET, and then release the BOOT button, at this time the module can enter the download mode to flash the firmware (demo) to solve the situation.

**Question: How to deal with the first compilation of the program being extremely slow?**

**Answer:**

- It's normal for the first compilation to be slow, just be patient

**Question: How to handle the display "waiting for download..." on the serial port after successfully ESP-IDF flashing?**

**Answer:**

- If there is a reset button on the development board, press the reset button; if there is no reset button, please power it on again

**Question: What should I do if I can't find the AppData folder?**

**Answer:**

- Some AppData folders are hidden by default and can be set to show.
- English system: Explorer->View->Check "Hidden items"
- Chinese system: File Explorer -> View -> Display -> Check "Hidden Items"



**Question: How do I check the COM port I use?****Answer:**

- Windows system:

- ①View through Device Manager: Press the Windows + R keys to open the "Run" dialog box; input devmgmt.msc and press Enter to open the Device Manager; expand the "Ports (COM and LPT)" section, where all COM ports and their current statuses will be listed.
- ②Use the command prompt to view: Open the Command Prompt (CMD), enter the "mode" command, which will display status information for all COM ports.
- ③Check hardware connections: If you have already connected external devices to the COM port, the device usually occupies a port number, which can be determined by checking the connected hardware.

- Linux system:

- ①Use the dmesg command to view: Open the terminal.
- ①Use the ls command to view: Enter ls /dev/ttyS\* or ls /dev/ttyUSB\* to list all serial port devices.
- ③Use the setserial command to view: Enter setserial -g /dev/ttyS\* to view the configuration information of all serial port devices.

**Question: Why does the program flashing fail when using a MAC device?****Answer:**

- Install MAC Driver ([https://files.waveshare.com/wiki/common/CH34XSER\\_MAC.7z](https://files.waveshare.com/wiki/common/CH34XSER_MAC.7z)) and flash again.

**Question: How to use SquareLine Studio to design interfaces?****Answer:**

- Please refer to SquareLine Studio tutorial ([https://www.waveshare.com/wiki/Waveshare\\_SquareLine\\_Studio](https://www.waveshare.com/wiki/Waveshare_SquareLine_Studio))

**Question: What is the maximum brightness of this product?**

**Answer:**

- Measured around 530 cd/m<sup>2</sup>.

# Support

## Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.

Working Time: 9 AM - 6 PM GMT+8  
(Monday to Friday)

Submit Now (<https://service.waveshare.com/>)

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(<https://www.waveshare.com/w/index.php?title=ESP32-S3-Touch-LCD-1.47&oldid=110303>)"*